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OBITUARY

WILLIAM SEALY GOSSET, 1876–1937

BOTH the practice and science of agriculture owe much to the efforts of men whose normal avocations are other than those of farming. Among these must be included William Sealy Gosset, whose passing the members of this Society, in common with many others, mourn to-day.

Born on the 13th June, 1876, Gosset was the eldest son of Colonel Frederic Gosset, late R.E., and of Agnes Sealy Gosset, both of whom survive him. He received his early schooling at Cullercoats and Gore Court, Sittingbourne, Kent, and later at Winchester. From there he proceeded to New College, Oxford, where he gained a 1st Class Mathematical Moderatorship in 1897, and a 1st Class in Natural Sciences in 1899.

From Oxford he joined the technical staff of Messrs. Arthur Guinness, Son & Co. in Dublin. In 1906 he married Marjory, the youngest daughter of the late James L. Phillpotts, Esq., for many years the Headmaster of Bedford Grammar School, who with one son and two daughters survive him.

The necessity produced the man in a very real sense in Gosset's case. The opening of the century witnessed an awakened interest in experimental agriculture and a more scientific approach to the improvement of crop plants. Amongst the many efforts proceeding from these influences was one instituted by the combined efforts of Messrs. Arthur Guinness, Son & Co. and the Irish Department of Agriculture to improve the Irish barley crop. The various schemes of experiment proceeded satisfactorily for a number of years, but gradually a position developed in which it became necessary to test smaller quantities of material than previously. In time the practical difficulties surrounding the cultivation of small plots of barley were surmounted, but the degree of reliability that could be attached to the results so obtained presented yet another and more serious difficulty.

I well remember the uneasiness with which I studied the figures of some of the earliest small-scale tests and the sense of relief when I was invited by a member of Messrs. Guinness' staff to submit my data to a "young member of the brewing staff named Gosset" for statistical examination. The letter I received in reply from Gosset, one of a long series which I still prize, was clear, concise, full of help and promise of future assistance—a key, as I learned in time, to a very sterling character.

In this way and from my data and those of other workers in the same field arose Gosset's appreciation of the practical and theoretical aspects of agricultural experimentation. His statistical treatment of samples derived from experiments which are inevitably limited in number has been amplified by later workers, but it is doubtful whether the position could have been advanced so rapidly had it not been for Gosset's ability to visualize practical limitations and issues. He displayed a readiness to work within limits prescribed by established conditions of varying nature, and thereby inspired confidence in those less familiar with the mathematical line of approach.

Then followed a series of papers with their authorship disguised under the *nom de plume* of "Student" and, arising out of them, requests for advice from all parts of the world. Thus, unheralded and unsought by himself, Gosset achieved an international reputation, and became the exponent of more scientific methods of agricultural experimentation.

Amongst his immediate colleagues at St. James's Brewery Gosset enjoyed an enviable reputation for the soundness of his views and clarity of exposition, and such was the confidence he inspired that no one was surprised when he was deputed to take control of Messrs. Guinness' new undertaking at Park Royal.

Gosset was the personification of modesty, courteous in manner to all, quiet in address, neither using dialectical arts himself nor being deceived by them in others. In argument he experienced no pleasure, although when necessary he would maintain his position with a delightfully disarming and very characteristic persuasiveness.

Those who were privileged to know Gosset in his home realized a pervading spirit of freedom and joyousness; there, discarding the weight of heavy responsibilities, he radiated that boyishness of spirit that made him at all times such a delightful but yet never flippant companion.

To his immediate friends Gosset leaves an example of loyalty and generosity in feeling and action, to a wider circle that of leadership in a subject he made peculiarly his own, but to all a sense of heavy loss.

H. H.

In the statistical world Gosset will be remembered above all for the notable contribution which he made to *Biometrika* in 1908. Concerned as he was with the problem of determining the significance of the mean of a sample from a normal population when the only knowledge regarding the standard deviation of the latter was the estimate derived from the sample, "Student" worked out the first four moments of the distribution of s^2 , the sample estimate of what was later called the variance, and fitted the appropriate Pearsonian

curve. This turned out, in fact, to be the exact solution of the problem. Proceeding to determine the random sampling distribution of $z = \bar{x}/s$, "Student" gave the test which is now imperishably associated with his name, and provided a short table for calculating the probability associated with a given value of z , later expanded and modified in "Metron" as the table of the t -distribution. The work of 1908 was the forerunner of the modern exact tests appropriate to small samples, which have proved to be of such inestimable value, particularly to the biologist.

This was far from being "Student's" only contribution to the theory of statistics, but it was a very notable single achievement. In particular his illustrations of what later became known as his method of paired differences provided an admirable example of the principle of reduction of error by the control of experimental design, and foreshadowed the extension of this principle to the now well-known experimental methods which were developed specifically in the realm of agricultural experimentation. One of the pioneer papers on the subject of the experimental error of field trials was that of Mercer and Hall in the *Journal of Agricultural Science* for 1911. To this paper "Student" contributed an appendix describing his method of utilizing a given area of land to the best advantage in testing two varieties. This method was used by E. S. Beaven in devising his well-known half-drill strip system for variety tests on cereals, described in an article in the *Journal of the Ministry of Agriculture* in 1922. "Student" himself contributed to *Biometrika* (Vol. 15) a paper dealing with Beaven's "chess-board" and half-drill strip methods, and more recently provided an admirable article on "Yield Trials" for the 1931 edition of Baillière's *Encyclopædia of Scientific Agriculture*.

Gosset became a Fellow of this Society in 1933, being one of the first to welcome with enthusiasm the creation of the Industrial and Agricultural Research Section, and he came to London specially to attend the inaugural meeting of the Section on November 23rd, 1933. On taking up his new duties in London he became a regular attender at the Section meetings, and on March 26th, 1936, he shed his anonymity and opened in person a discussion on "Co-operation in Large-Scale Experiments," which ranged over the whole field of his own contributions to the subject, and led to a most interesting and valuable series of communications from subsequent speakers.

J. W.

An old friend (E. M. E.) writes :—

"Alas, I cannot put my hand on one or two letters I know I kept—one about his hat, the only one he ever possessed, I believe. He came into my room at University College flourishing a hat, and

explained that he had bought it to go to a Board meeting in town—under family pressure, I suspect. He wrote soon after, and confessed that he had left it in the train, and that his son would have to retrieve it from the Lost Property Office.

“Did you ever hear Gosset say an unkind thing about anyone? He had an excuse for all the failings of other people, and how he enjoyed life—wet or fine—in bad days and good! It is indeed a tragedy that he should be deprived of life at 61, and his friends of one who made for peace and happiness.

“When I begin to think of Gosset, I don’t think I saw anything much of him till after the war, when he used to come to the Lab. on his yearly holidays for a passing visit and later as an examiner; then he generally came straight to my room, dumped his rucksack in my arm-chair, and asked whether ‘himself’ was free; he would have tea in the common room, but seldom talked work. At one time he was great on crossing loganberries and blackberries (I think this was the combination), and he got something he called a jam-berry, and told us how to make jam of his berry (you must not boil it), and brought up a pot which we added to our bread and butter at tea-time. Generally he and I had a short chat, and others used to claim his company at tea.

“I doubt whether I knew him very well, but our attitude was similar, and we saw things from the same point of view, but we did not correspond except once or twice about work. He came here with one of his daughters a year ago last month, and we talked of old times and present difficulties, and I was struck as I always was by his enjoyment of what I should call the simple pleasures of life.”

PROFESSOR GEORGE WILLIAM DANIELS

It is hard to believe that Professor Daniels is dead. He died from heart trouble on December 17th, 1937—a few weeks before his sixtieth birthday. He had been for so long an essential part of the academic and business life of Manchester—belonging to Manchester and to nowhere else—that it is difficult to think of Manchester without him. It is sad to reflect that he will never again be seen wearing what always seemed to us the same light grey suit, the same heavy brown shoes and, in winter, the same macintosh, walking across the quadrangle regularly at nine o’clock each morning. Collecting his letters from his box in the porter’s office, he used to climb the steps and make his way along the corridor to his room in the farthest corner of the main building of Owens College overlooking Burlington Street. There was his sanctum—a square room with a rolled-top desk with a portrait of Marshall hanging above, untidy piles of *Punch* sharing the window-ledge with a telephone (“I’ll have that